Attention and Transformers

ECON 8310: Business Forecasting — Lecture 9, Part 2

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October 13, 2026

Lecture Outline

1 What Recurrence Could Not Do

2 The Attention Mechanism

3 The Transformer Encoder

4 Transformers for Time Series

5 Key Takeaways

What Recurrence Could Not Do

Two constraints that are built into reading a sequence one step at a time.

Two Problems Built Into Recurrence

Part 1's recurrent forecaster has two problems left, both from reading strictly left to right.

The information bottleneck. Everything about step 3 must squeeze through every hidden state between 3 and T. Gates help, but the whole history still lives in one fixed-size vector.

The sequential bottleneck. Step t waits for step $t-1$: a 1,000-step sequence takes 1,000 sequential operations, however many GPUs you own.

> **Attention removes both at once.** Every position looks directly at every other, in one operation: no compression through intermediate states, no ordering constraint on the computation.

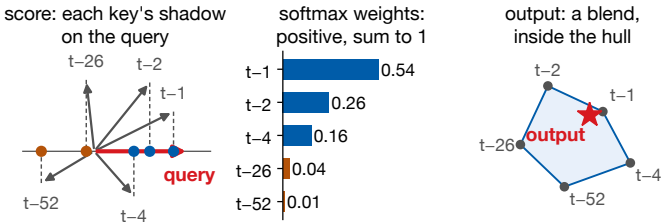
The Attention Mechanism

A learned, weighted lookup — which is a more useful way to read it than the formula suggests.

Scaled Dot-Product Attention (Vaswani et al. 2017)

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V$$

A lookup with soft matching: each week's **query** is scored against every week's **key**, and the scores weight the **values**.

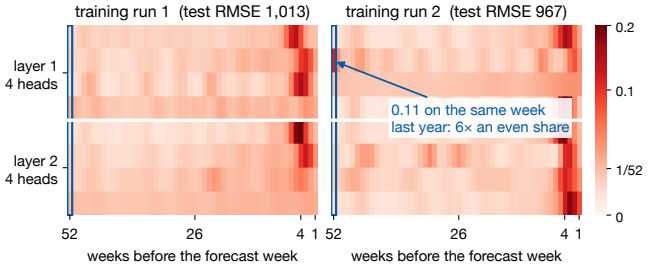


a two-dimensional cartoon of one 16-dimensional head

The score is the query's length times the key's shadow on it. Softmax makes the weights positive and sum to one, so the output stays *inside* the values: attention averages, it cannot extrapolate. The $\sqrt{d_k}$ keeps the softmax from saturating.

What That Buys a Forecaster

Attention can weight any week in the window in one step; an RNN needs 52 updates to reach last year. Where our trained model actually looked, over the test year:



Mostly the last two to seven weeks; in one of two training runs, one head found last year's week. The weights are **learned, not specified** — and they show where the model looked, not what mattered.

> The price is quadratic: $O(T^2)$ in time and memory — fine at $T = 52$, painful at $T = 10{,}000$.

Multi-Head Attention

One attention operation gives one set of weights — one notion of what is relevant. **Multi-head attention** runs h of them in parallel on different learned projections, then concatenates:

$$\text{MultiHead}(Q, K, V) = \text{concat}(\text{head}_1, \ldots, \text{head}_h)\, W^O$$

Each head can specialize. Ours did: most watch the last few weeks, one spreads its weight evenly, and in one run one found the annual lag.

The cost is not what you would guess. Each head works in d_{model}/h dimensions, so eight heads cost about the same as one full-width head.

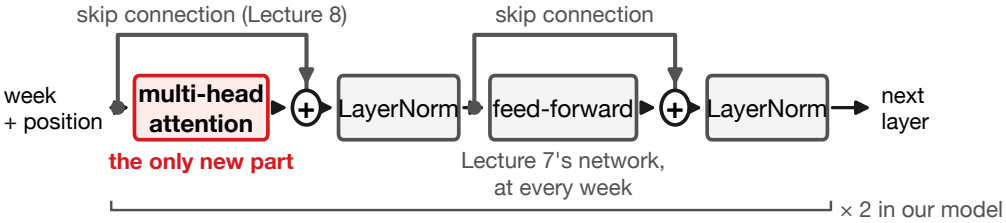
Our model: `nhead=4, d_model=64` — four heads of 16 dimensions.

The Transformer Encoder

Attention, plus the three pieces that make it trainable.

The Encoder Block

A Transformer encoder layer is four components in a fixed arrangement:



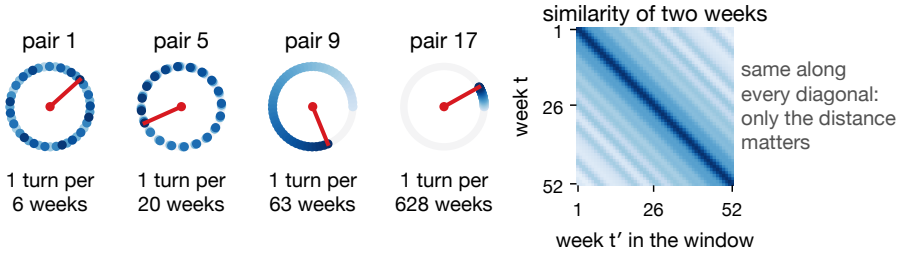
Stack N of those and you have the encoder. Nothing in the block is new to you except attention itself — the skip connections are ResNet's, LayerNorm rescales each week's vector, and the feed-forward sublayer is the Lecture 7 network applied to every week.

One thing is conspicuously missing, and it is not an oversight. Nothing above depends on the *order* of the input.

Positional Encoding: Attention Cannot Tell Time

Attention is permutation-invariant: shuffle the window and every output is unchanged. A forecaster that cannot tell last week from thirty weeks ago is not forecasting.

The fix adds 32 pairs $(\sin \omega_i t,\ \cos \omega_i t)$, $\omega_i = 10000^{-2i/d}$, to each week — each pair a clock hand turning at its own speed:



Moving Δ weeks turns every hand by its own fixed angle, so how alike two weeks look depends only on their distance. **We measure what removing it costs at the end.**

Why Transformers Took Over

Three properties, in the order they mattered commercially.

Parallelism. Every position is computed at once, so training scales across hardware. More than accuracy, this is why it won.

Direct long-range paths. Any position reaches any other in one step.

Scale. It kept improving as parameters and data grew. BERT (Devlin et al. 2019) is an encoder stack for understanding, GPT a decoder stack for generation — both this block, repeated.

> Every one of those advantages is about **large data and long sequences**. Hold that thought against a 52-week window and 5,160 training examples.

Transformers for Time Series

The implementation, the measurement, and an honest reading of it.

Transformer Encoder in PyTorch

Minimal forecasting encoder

```
layer = nn.TransformerEncoderLayer(d_model=64, nhead=4,
dim_feedforward=128, dropout=0.1, batch_first=True)
self.enc = nn.TransformerEncoder(layer, num_layers=2)
self.inp = nn.Linear(n_features, 64)
self.fc  = nn.Linear(64, 1)

def forward(self, x):
h = self.inp(x) + self.pe     # position must be added
return self.fc(self.enc(h)[:, -1, :]).squeeze(-1)
```

`nn.TransformerEncoderLayer` gives you the whole block — but not the positional encoding. You add that yourself, and forgetting is a silent failure.

Multi-step forecasting also needs a causal mask.

Does It Work?

Same weekly panel and test block as everything since Homework 3; every sequence model reads 52-week windows.

Model	Test RMSE	Parameters
LASSO, 46 engineered features	**744**	46
XGBoost	781	—
Vanilla RNN	**805**	4,545
Random forest	899	—
Transformer encoder, 2 layers	**990**	67,329
LSTM	998	17,985
1D CNN, best variant (26-week window)	1,032	8,161
Seasonal naive	2,152	0

The Transformer beats the CNN and **ties the LSTM** — 990 against 998, a difference far smaller than the run-to-run spread. It loses to a vanilla RNN holding **one-fifteenth** as many parameters, and to a 46-coefficient linear model.

Reading That Result Honestly

Every advantage on the "why Transformers took over" slide is an advantage *at scale*, and this problem has none of the scale.

Parallel computation	Real, but irrelevant — the RNN trains in seconds either way.
Direct long-range paths	Worth little when the signal is dominated by recent weeks.
Scales with data	5,160 training windows is not the regime where that pays.

And positional encoding is doing real work, which we can measure. Remove it and the same model must forecast from an unordered bag of 52 weeks:

Transformer **with** positional encoding	**990**
Transformer **without** positional encoding	1,282

292 RMSE — dropping it below even the 1D CNN. That gap is the price of permutation invariance, and the reason the sinusoids exist.

Key Takeaways

Six architectures, one series, and what the ranking actually taught us.

Sequence Architecture Comparison

Architecture	Long range	Parallel	Needs
1D CNN	receptive field only	yes	modest data
Vanilla RNN	$\sim$10–30 steps	no	modest data
LSTM	hundreds of steps	no	more data
Transformer	unlimited, $O(T^2)$	**yes**	**much more data**

Choose by the structure of your problem, not by recency of the architecture. Short window and recent-lag signal: a plain RNN, or honestly a linear model. Long sequences, abundant data, dependencies at arbitrary distance: a Transformer, and nothing else comes close.

That second case is language, which is why this architecture reshaped that field and has been slower to displace gradient boosting in tabular forecasting — where the M5 competition was won by ensembles of exactly the methods in Lectures 5 and 6.

Lecture 9 Part 2: Key Takeaways

1. **Attention**: score queries against keys, softmax, average the values. Distance costs nothing.
2. The price is **quadratic** in sequence length: fine at 52 steps, the central constraint at 10,000.
3. **Multi-head** attention runs several notions of relevance at roughly the cost of one.
4. An encoder block: attention, **skips**, LayerNorm, feed-forward. Only attention is new.
5. **Attention is permutation-invariant**: add position yourself — PyTorch will not, and forgetting fails silently.
6. Here it **tied the LSTM** and lost to an RNN with **one-fifteenth** the parameters: its advantages are advantages at scale.

Next: Bayesian Statistics I — a different answer to what a forecast should tell you.

References I

- Devlin, Jacob et al. (2019). "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding". In: *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics*, pp. 4171–4186.
- Vaswani, Ashish et al. (2017). "Attention Is All You Need". In: *Advances in Neural Information Processing Systems*. Vol. 30, pp. 5998–6008.